Scale-Conditioned Geometry of the Wildland–Urban Interface: A Theory of Anthropogenic–Ecological Boundaries

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Abstract

Boundary length is widely used to characterize exposure in the wildland urban interface (WUI), yet it is often reported as if it were a fixed geographic property of a landscape. This interpretation overlooks a fundamental geometric fact. The length of any mapped boundary depends on both how the boundary is defined and the scale at which it is measured. In this paper we develop a scale conditioned framework for interpreting WUI boundary length that separates two independent sources of variation. The first is definition scale, which determines the boundary object through classification choices such as settlement representation, vegetation thresholds, neighborhood radius, and adjacency rules. The second is measurement scale, which determines the estimated perimeter of that object through geometric resolution. Drawing on scaling theory from fractal geometry and wildfire perimeter analysis, we formalize WUI boundary length as a family of functions L_d(ε), where d indexes defensible delineation bundles and ε indexes measurement scale. We then interpret WUI boundaries as generative anthropogenic ecological interfaces produced by interacting spatial processes including infrastructure mediated settlement expansion, vegetation mosaic dynamics, and wildfire disturbance. Under this view, WUI boundary geometry encodes information about coupled human and ecological pattern formation. The principal implication is that apparent disagreement among reported WUI boundary lengths is often an expected consequence of scale specification rather than contradictory empirical evidence. Recasting WUI boundary length as a scale conditioned estimand provides a conceptual foundation for transparent wildfire risk analysis, cross study comparison, and improved interpretation of anthropogenic landscape edges.

Keywords: wildland–urban interface; wildfire; boundary scaling; fractal geometry; anthropogenic–ecological systems; geospatial inference

1. Introduction

The wildland urban interface represents one of the most consequential spatial configurations in wildfire science (Radeloff et al., 2005, 2018). Where settlements intermix with flammable vegetation, fire behavior intersects directly with the built environment and produces coupled socio ecological risk systems (Moritz et al., 2014, Bar-Massada et al., 2014). These systems influence land management, infrastructure planning, and public safety (Cohen, 2000, Calkin et al., 2014). As development expands into vegetated landscapes, the spatial extent and geometry of the wildland urban interface have become central concerns for wildfire mitigation and climate adaptation (Carlson et al., 2022, Radeloff et al., 2018).

Despite the prominence of WUI mapping, one of the most frequently reported derived quantities remains poorly interpreted (Bar-Massada, 2021, Hanberry, 2020). Boundary length is often treated as a single scalar property of a landscape (Stewart et al., 2007, Bar-Massada, 2021, Hanberry, 2020). This interpretation implicitly assumes that the boundary is uniquely defined and that its length can be measured independently of analytical assumptions. Neither assumption is generally valid. Boundary length is not an intrinsic geographic constant. It is an estimand whose value depends on both how the boundary is constructed and how it is measured (Stewart et al., 2007, Mandelbrot, 1967, Richardson, 1961).

Two distinct mechanisms generate this dependence. The first arises during boundary construction: classification choices regarding settlement representation, threshold values, neighborhood radius, and adjacency rules produce different WUI geometries (Stewart et al., 2007, Carlson et al., 2022). The second arises during boundary measurement: even for a fixed geometric object, estimated perimeter varies with measurement resolution, consistent with coastline-paradox and fractal-scaling logic (Mandelbrot, 1967, Richardson, 1961). These mechanisms are conceptually distinct but are often conflated in interpretation.

Recognizing this structure changes the scientific question. Instead of asking for one perimeter value, the relevant question is how interface geometry behaves across measurement scales under alternative delineation assumptions. Beyond measurement, the WUI boundary is best interpreted as an emergent anthropogenic–ecological interface generated by interacting spatial processes. Settlement expansion follows infrastructure networks and economic gradients, vegetation patterns emerge from ecological heterogeneity and disturbance history, and wildfire both responds to and reshapes this interface (McAlpine and Wotton, 1993, Hargrove et al., 2012, Macauley et al., 2022, Bar-Massada et al., 2014, Cadenasso et al., 2003).

This study develops a framework for interpreting wildland urban interface boundary length under explicit scale dependence. First, we formalize WUI boundary length as a scale conditioned family of functions defined across alternative delineation bundles and measurement scales. Second, we distinguish between definition scale and measurement scale as two independent sources of variation in reported perimeter values. Third, we interpret WUI boundaries as generative anthropogenic ecological interfaces produced by interacting spatial processes including settlement expansion, veg-

etation mosaic formation, and wildfire disturbance. Together these elements provide a coherent conceptual foundation for interpreting WUI boundary geometry in both scientific analysis and decision support contexts.

2. Scale Dependence in Boundary Geometry

Irregular boundaries rarely possess a single invariant length. The coastline paradox demonstrates that measured length increases as ruler size decreases because finer measurements capture additional spatial irregularity (Mandelbrot, 1967, Richardson, 1961, Turcotte, 1997). In this setting, boundary length is better described as a scaling relationship than as a single scalar.

A common representation is

$$L(\varepsilon) = c\,\varepsilon^{1-D},$$

where $L(\varepsilon)$ is measured length at scale ε, c is a proportionality constant, and D is an effective roughness dimension. In logarithmic form,

$$\log L = \log c + (1 - D)\log \varepsilon,$$

In this representation the parameter D summarizes the effective geometric roughness of the boundary across the analyzed scale interval. When D approaches one the boundary behaves approximately like a smooth Euclidean curve. As D increases above one the boundary becomes increasingly irregular and exhibits properties associated with fractal geometry. This interpretation motivates slope based diagnostics over defined scale ranges rather than attempts to estimate a single perimeter value (Mandelbrot, 1967, Hargrove et al., 2012, Turcotte, 1997).

Wildfire perimeter studies provide direct precedent. Perimeter geometry reflects interacting controls (fuels, weather, topography), and observed fractal-like complexity changes with analysis scale (McAlpine and Wotton, 1993, Hargrove et al., 2012, Macauley et al., 2022). These findings indicate that irregular boundaries can encode process information rather than merely measurement error.

The same principle applies to WUI. Settlement structure intersects heterogeneous vegetation mosaics, producing boundaries with multi-scale irregularity. Consequently, perimeter estimates are inherently scale-conditioned and should be interpreted through curves and diagnostics rather than single values (Radeloff et al., 2005, Stewart et al., 2007, Carlson et al., 2022).

3. Definition Scale and Measurement Scale

The scale dependence of WUI boundary length arises from two independent mechanisms (Stewart et al., 2007, Carlson et al., 2022, Richardson, 1961).

3.1. Definition Scale

Definition scale refers to assumptions embedded in delineation rules. Analysts choose settlement representation (e.g., housing-density aggregates versus building-level points), development and vegetation thresholds, neighborhood radius, adjacency criteria, and urban exclusions (Radeloff et al., 2005, Stewart et al., 2007, Carlson et al., 2022). These choices alter the mapped WUI geometry and therefore alter the boundary object itself.

3.2. Measurement Scale

Measurement scale refers to the geometric resolution used to estimate perimeter for a fixed boundary object. Smaller rulers or finer geometric approximations capture greater local sinuosity and generally return longer estimates (Mandelbrot, 1967, Richardson, 1961). Thus, perimeter variation with ε persists even when delineation assumptions are held constant.

3.3. Formal Decomposition

Let $d \in \mathcal{D}$ denote a defensible delineation bundle, and let B_d denote the resulting boundary object. Measured boundary length at geometric scale ε is

$$L_d(\varepsilon) = \mathcal{M}(B_d, \varepsilon),$$

where $\mathcal{M}$ is a specified measurement operator. Under scale-structured geometry, an interval-specific approximation is

$$L_d(\varepsilon) = c_d\, \varepsilon^{1-D_d}.$$

The estimand is therefore not a scalar but the family $\{L_d(\varepsilon) : d \in \mathcal{D}\}$. Scientifically defensible interpretation requires reporting both object variation across d and measurement variation across ε (Carlson et al., 2022, Bar-Massada, 2021, Hanberry, 2020).

4. Generative Processes of the WUI Boundary

Wildland urban interface boundaries do not arise solely from geometric partitioning of space. They emerge from the interaction of multiple spatial processes operating across human and ecological systems. Settlement expansion typically follows infrastructure networks and economic gradients rather than spreading uniformly across space. Vegetation landscapes exhibit heterogeneous mosaics shaped by environmental gradients, disturbance history, and ecological succession. When development intersects these mosaics the resulting boundary inherits structural complexity from both anthropogenic and ecological pattern formation (Bar-Massada et al., 2014, Cadenasso et al., 2003, White and Engelen, 1993, Batty, 2008, Turner, 1989).

Settlement expansion is typically network-mediated rather than spatially uniform. Transportation corridors, zoning regimes, economic gradients, and topographic constraints channel development into branching or clustered configurations that propagate through infill and leapfrog dynamics (Bar-Massada et al., 2014, Li et al., 2022, White and Engelen, 1993, Batty, 2008).

Vegetation landscapes exhibit heterogeneous mosaics driven by environmental gradients, disturbance history, and ecological succession. Development advancing into these mosaics intersects patch boundaries and increases interface irregularity (Radeloff et al., 2018, Bar-Massada et al., 2014).

Wildfire introduces an additional coupling mechanism. Fire behavior responds to fuel and weather structure while also reorganizing ecological pattern through disturbance; post-fire regeneration and rebuilding can subsequently reshape boundary complexity. In this sense, wildfire both interacts with and reorganizes anthropogenic–ecological edges (Hargrove et al., 2012, Macauley et al., 2022).

The combined effect is a boundary with structure across multiple scales: infrastructure diffusion contributes branching anthropogenic form, vegetation mosaics contribute ecological irregularity, and wildfire disturbance contributes temporal reorganization of both (McAlpine and Wotton, 1993, Hargrove et al., 2012, Macauley et al., 2022, Bar-Massada et al., 2014).

5. A Minimal Dynamical Representation

To illustrate mechanism-level interpretation, consider settlement density $S(x,t)$, vegetation biomass $V(x,t)$, and wildfire activity $F(x,t)$ over spatial position x and time t.

The following system is not intended as a predictive model of WUI dynamics. Instead it provides a minimal conceptual representation that illustrates how interacting settlement, vegetation, and fire processes can generate spatial interfaces with complex geometry. In this view the WUI boundary corresponds to spatial contours where settlement and vegetation fields intersect under chosen classification thresholds (Moritz et al., 2014, Liu et al., 2007).

A minimal coupled system is

$$\partial_t S = D_s \nabla^2 S + f(S, I),$$

$$\partial_t V = D_v \nabla^2 V + g(V) - h(S, V, F),$$

$$\partial_t F = k(V, W, T),$$

where D_s and D_v are diffusion-like parameters, I denotes infrastructure and development drivers, W denotes weather forcing, and T denotes topographic controls.

In this representation, WUI boundaries are contours where settlement and vegetation fields intersect under chosen thresholds. Although simplified, the system clarifies that observed interface geometry can be interpreted as the spatial trace of coupled human–environment dynamics rather than as a

static cartographic artifact (Liu et al., 2007, Cadenasso et al., 2003).

6. Implications for WUI Analysis and Decision Support

This framework clarifies several recurrent interpretation problems.

First, differences in reported WUI boundary length across studies are often expected consequences of different delineation bundles or measurement scales rather than contradictory findings. Comparability improves when studies report full d definitions, scale grids for ε, and diagnostics for the analyzed scaling interval (Bar-Massada, 2021, Hanberry, 2020).

Second, WUI boundaries should be treated as dynamic features that evolve with settlement change, ecological succession, and disturbance. Longitudinal monitoring of boundary geometry can therefore provide information about changing anthropogenic–ecological coupling (Radeloff et al., 2018, Macauley et al., 2022, Bar-Massada et al., 2023).

Third, WUI can be interpreted as one member of a broader class of anthropogenic edge systems (e.g., agricultural frontiers, forest-fragmentation edges, peri-urban transitions). This link opens methodological transfer between wildfire science and landscape-process theory (Bar-Massada et al., 2014, 2023, Cadenasso et al., 2003).

For rigorous reporting, a minimum disclosure set should include: (i) complete delineation bundle d, (ii) measurement protocol and scale interval, (iii) perimeter-by-scale curves $L_d(\varepsilon)$, (iv) log–log diagnostics, and (v) sensitivity of conclusions across plausible alternative delineation bundles (Carlson et al., 2022, Bar-Massada, 2021, Hanberry, 2020).

7. Conclusions

The length of the wildland urban interface boundary is not a fixed geographic property of landscapes. It is a scale conditioned estimand whose value depends jointly on how the boundary is defined and how it is measured. Recognizing this structure reframes WUI perimeter estimation from a simple measurement problem into a formal inference problem that requires explicit specification of delineation assumptions and measurement scale (Stewart et al., 2007, Mandelbrot, 1967, Richardson, 1961).

Interpreting WUI geometry also requires a generative perspective: settlement expansion, vegetation mosaics, and wildfire disturbance jointly produce anthropogenic–ecological boundaries with multiscale structure (Moritz et al., 2014, Hargrove et al., 2012, Macauley et al., 2022, Bar-Massada et al., 2014). Representing perimeter as $L_d(\varepsilon)$ provides a coherent framework for this interpretation, supports transparent cross-study comparison, and strengthens decision support for wildfire risk analysis, planning, insurance analytics, and policy reporting.

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Conflicts of Interest

The authors declare no conflict of interest.

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